

Dr. Joy Bose

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Professional Summary

Senior ML/AI systems leader specialising in LLM-based architectures — RAG pipelines, knowledge graphs, and MCP-based agentic systems — with a focus on building production-grade systems that remain reliable beyond prototype scale. Combines deep research foundations (PhD in neuromorphic computing, University of Manchester; MSc in Neuroscience, King's College London) with 15+ years of real-world deployment across telecom, edtech, and browser intelligence. Particularly strong at identifying where current AI approaches break and designing systems that hold up in practice. Co-inventor on 8 granted US and European patents; 20+ peer-reviewed publications; winner of IP awards at Samsung and delivery awards at Ericsson. GitHub: github.com/joybosero

Key Credentials at a Glance

• PhD, Computer Science – Univ. of Manchester • MSc, Psychology & Neuroscience – King's College London • LLM, International Business Law – Golden Gate Univ. (Summa Cum Laude, 4.0) • BE (Hons, 1st Class) Computer Science – MNNIT Allahabad	• 8 granted US/European patents; 20+ filed • 20+ peer-reviewed papers; 707 citations, h-index 13 (Google Scholar) • Awards: Ericsson Power Award, Samsung Star IP, Best Paper x2 • 15+ yrs ML/AI; 5+ yrs LLMs, RAG, agentic systems
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Technical Skills

AI / ML: LLMs, RAG, Agentic systems (MCP), Knowledge Graphs, LLMops, Prompt Engineering, Transformers, LangChain, Spiking Neural Networks, SVM, FastText, Recommender Systems, Supervised/Unsupervised learning

Cloud & MLOps: Azure ML, AWS SageMaker, Amazon Q, OpenSearch

Languages: Python (10+ yrs), Java (10+ yrs), C (15+ yrs), C++ (2 yrs), MATLAB (5 yrs)

Domains: Telecom (BSS/OSS), EdTech, Browser Intelligence, Neuroscience & BCI, Logistics

Work Experience

Senior Data Scientist | Ericsson Global, Bengaluru

Jul 2020 – Present

- Built NLP-to-SQL system with RAG for telecom data access, eliminating manual query bottlenecks for multiple teams. Developed LLM-based automated Java code review pipeline (open-source Llama models) adopted by ~400 engineers across the BCSS organisation, saving an estimated ~20 minutes per review cycle.
- Designed knowledge graph + LLM solution (FalkorDB, MCP-based) for intelligent code understanding and log querying via OpenSearch; system adopted by 1,000+ engineers across multiple BCSS teams.
- Applied ML to identify inefficient radios for energy savings, predict trouble ticket resolutions, and categorise financial variance for CSP clients.
- Conducted comparative ML cost analysis across Azure and AWS; published Ericsson blog on AI-driven BSS financial data categorisation.
- 1 granted, 3 patents filed (EPO/WIPO); 6 papers published including 1 journal paper (IJMEMS 2025).
- Awards: **Power Award (Individual) 2022; Act to Accelerate Award (Team) 2021; AllStars Gold Award 2024.**

Lead Data Scientist | Jio Embibe, Bengaluru

Jan 2020 – Jun 2020

- Built ML models to analyse and improve student performance in standardised exams; developed nudge generation system for personalised learning outcomes.
- Paper published at EDM 2020 on auto-generation of diagnostic assessments.

Senior Software Development Engineer | Microsoft India (Edge Team), Hyderabad *Aug 2018 – Dec 2019*

- Trained ML model for relevant image extraction in Edge reading view using Bing's Virtual DOM; built client-side neural network for web form field classification (autofill).
- 1 US patent granted; 1 paper at IEEE INDICON 2019; presented at Microsoft MLADS SYNAPSE 2019.

Senior Chief Engineer, Data Intelligence Group | Samsung R&D, Bengaluru *Nov 2011 – Aug 2018*

- Led 4-member team on calendar event classification and personalised recommendation system with 94% accuracy across English and Korean.
- Research PoCs: EEG attention mapping to web DOM elements (Neurosky BCI); real-time news bias detection; web push notification CTR prediction; gesture-based SMS encryption via heteroassociative neural networks.
- Mentored interns, freshers, and first-time inventors; taught internal Data Structures and Algorithms courses.
- Awards: **Star IP Award 2013; Best Paper x2 (INDICON 2015, IEEE DISCOVER 2016); Tech Paper Shepherd Award 2017; Top 10 finalist, C-Lab Intrapreneurship 2017.**

Software Engineer | Digital Applications International (now Dematic), Stockport, UK *Oct 2007 – Oct 2011*

- Software development for logistics Warehouse Management Systems (Matflow) for clients including Asda/Walmart, Screwfix, B&Q, Lever Fabergé.

Education

PhD, Computer Science (Spiking Neural Networks) | University of Manchester, UK *2002–2007*
Overseas Research Studentship (3 years). 3 international + 3 UK conference papers + 1 IEEE Journal paper.

MSc, Psychology and Neuroscience of Mental Health (Merit) | King's College London, UK *2021–2023*
Dean's Award for Outstanding Student Performance in Mental Health in the Community.

LLM, International Business and Finance Law — Summa Cum Laude, GPA 4.0 | Golden Gate University, USA *2024–2025*

BE (Hons, 1st Class), Computer Science — 82.04% aggregate | MNNIT Allahabad, India *1998–2002*
Merit scholarship (3 years); NTSE Scholarship; TCS scholarship (HiPC 2001).

Selected Publications

1. Bose, J. and Adyapak, S. A Survey of Security, Privacy, and Ethics of Large Language Models in Telecom. COMSNETS 2026, Bengaluru, pp. 1100–1104.
2. Ramesh S., Bose J. et al. Automated Code Review Using Large Language Models at Ericsson: An Experience Report. ICSME 2025, pp. 602–607.
3. Bandyopadhyay, S., Bose, J. et al. A Hybrid Framework for Real-Time Data Drift and Anomaly Identification Using Hierarchical Temporal Memory. IJMEMS 10(3), 2025.
4. Pardhasaradhi N., Bose J. et al. Identification of Inefficient Radios for Efficient Energy Consumption in a Mobile Network. COMSNETS 2024.
5. Bose J. et al. Prediction of Throughput Degradation from Trouble Frequencies, given Environmental Unknowns. COMSNETS 2022.
6. Dhavala S., Bose J. et al. Auto Generation of Diagnostic Assessments and Their Quality Evaluation. EDM 2020.
7. Singla K., Bose J., Varshney N. Word Embeddings for IoT Based on Device Activity Footprints. Computación y Sistemas 23(3), 2019.
8. Patankar A., Bose J. A Bias Aware News Recommendation System. ICSC 2019.
9. Bose J. et al. A Hands Free Browser Using EEG and Voice Inputs. IJCNN 2015.
10. Furber S.B., Brown G., Bose J. et al. Sparse Distributed Memory Using Rank-Order Neural Codes. IEEE Trans. Neural Networks 18(3), 2007.

Granted Patents

- EP3750115B1 — Machine learning on a blockchain
- US11295492B2 — Electronic device related to rendering of web content
- US11030448B2 — Method for recommending one or more actions (mobile device)
- US10235587B2 — System for optimizing image capturing boundary
- US10210598B2 — Electronic device for displaying and processing images
- US9992254B2 — Method for providing a web feed in a URI
- US12591731B2 — System and Method for Selecting Relevant Content in an Enhanced View Mode
- EP4331295B1 — Reducing Interference in a Communications Network

Complete list available on Google Scholar and Google Patents. Full patent list available on request.

Other

Volunteering: NIMHANS Bengaluru certified wellbeing volunteer.

Interests: Google Local Guides India (Level 10); AI ethics; neuroscience-AI intersection; contemplative philosophy.